

The $(1\rho) \times (1\rho)$ piece of $|\mathfrak{B}_2|^2$

Transverse divergences, the DGLAP question, and the p^+ integration

$A \equiv (x - z)^2$, $B \equiv (x - w)^2$, $C \equiv (z - w)^2$, $A' \equiv (x' - z)^2$, $B' \equiv (x' - w)^2$, $C' \equiv (z - w')^2$; $S \equiv p^+ + k^+$, $\zeta \equiv k^+/S$, $\bar{\zeta} \equiv p^+/S$; $\vee \equiv q^+$ the total $+$ -momentum, Λ the rapidity cutoff on p^+ .

Summary. The prefactor and the colour structure are right (three bookkeeping corrections in §1). **There is no transverse UV divergence anywhere** — all 26 collapsing subsets of $\{x, x', z, w, w'\}$ are convergent (§3), including the "loop closing" region $z \rightarrow w$. **There is exactly one transverse IR divergence**, from $|z| \rightarrow \infty$, and after the p^+ integration its coefficient is remarkably clean (§6):

$$\frac{1}{4k^+} \log \frac{\vee - k^+}{\Lambda} \frac{(x - w) \cdot (x' - w')}{(x - w)^2 (x' - w')^2} \int \frac{d^2 z}{|z|^2},$$

i.e. (LO transverse structure) $\times$ (full rapidity span), with the z -gluon completely decoupled. **That is a soft / rapidity log, not a collinear one: this row gives no DGLAP** (§7) — the would-be collinear region $z \rightarrow w$ is only ρ^{-1} singular, and the ζ -weight of the log is $1/(\zeta\bar{\zeta})$, which is *part* of P_{gg} but not P_{gg} . The complete p^+ -integrated answer is boxed in §5.

1. The expression as written

1.1 Prefactor — correct

With $\mathfrak{B}^{(1)} = \frac{ig^2 \sqrt{p^+ k^+}}{4\pi^2 S}$ and the standard $\frac{4}{(2\pi)^3}$,

$$\frac{4}{(2\pi)^3} \mathfrak{B}^{(1)} \mathfrak{B}^{(1)\dagger} = \frac{g^4}{32\pi^7} \frac{p^+ k^+}{S^2} = \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{p^+ k^+}{S^2},$$

exactly your quoted coefficient (ratio = 1 symbolically). The vertex's own $1/S$ is kept *inside* the brackets, which is why they carry S/k^+ and S/p^+ — consistent.

1.2 Colour — correct as written

Relabelling the source index $a \rightarrow d$, the ket is $U^{ac}(w) f^{cbd} \rho^d(x) - U^{bc}(z) f^{acd} U^{de}(x) \rho^e(x)$, exactly as you have it. The two terms carry *different* structure constants (f^{cbd} vs f^{acd}), so f is correctly **not** pulled out front.

Three bookkeeping items.

1. The bra's second term reads $f^{ac'd'} U^{bc'}(z) U^{e'd'}(x') \rho^{e'}(x')$ while the ket's reads $f^{acd} U^{bc}(z) U^{de}(x) \rho^e(x)$: the two Wilson-line indices are **transposed** relative to each other. Numerically $\max |U(x) - U(x')^T| = 1.03$ on a random configuration, so this is not a harmless relabelling — fix the ordering to be uniform.
2. The measure should be $\int_{x, x', z, w, w'}$, not $\int_{z, w, w'}$: both x and x' appear, in the kernels and in $\rho^d(x)$, $\rho^d(x')$.
3. $|\mathfrak{B}_2|^2$ has **three** pieces. This row is $(1\rho) \times (1\rho)$; the cross terms $(1\rho) \times (2\rho)$ are your N_1 ; the $(2\rho) \times (2\rho)$ piece, coefficient $\frac{1}{(2\pi)^3} \frac{g^4}{16\pi^4} \frac{1}{p^+ k^+}$, is still missing. It matters for §6, because it carries the same $1/|z|^2$ tail.

2. The vertex factorises — this is what makes everything below tractable

Write the bracket as $\mathcal{W}^{ji}(x, z, w)$. It is *not* a generic 2×2 tensor: each of the two non- δ structures is an outer product of one fixed vector with one composite vector,

$$\boxed{\mathcal{W}^{ji} = \delta^{ji} \frac{A - B}{2C} + \frac{S}{k^+} (x - z)^j \mathcal{V}_w^i + \frac{S}{p^+} \mathcal{V}_z^j (x - w)^i}$$

$$\mathcal{V}_w^i \equiv \frac{(z - w)^i}{(z - w)^2} - \frac{(x - w)^i}{2(x - w)^2}, \quad \mathcal{V}_z^j \equiv \frac{(z - w)^j}{(z - w)^2} + \frac{(x - z)^j}{2(x - z)^2}.$$

(Reconstruction agrees with the coded vertex to 7×10^{-15} .) Call the three structures $\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3$ in that order; the primed ones use x', w' and the **same** z . Then $\mathcal{W}^{ji}\mathcal{W}^{ji}$ has exactly seven contractions, all reducing to dot products of 2-vectors:

contraction	closed form	closed	direct
$\mathcal{T}_1 \cdot \mathcal{T}'_1$	$\frac{(A-B)(A'-B')}{2CC'}$	-34.816293107	-34.816293107
$\mathcal{T}_1 \cdot \mathcal{T}'_2$	$\frac{A-B}{2C} (x' - z) \cdot \mathcal{V}_{w'}$	22.097763963	22.097763963
$\mathcal{T}_1 \cdot \mathcal{T}'_3$	$\frac{A-B}{2C} \mathcal{V}'_z \cdot (x' - w')$	13.171506168	13.171506168
$\mathcal{T}_2 \cdot \mathcal{T}'_2$	$[(x-z) \cdot (x'-z)] [\mathcal{V}_w \cdot \mathcal{V}_{w'}]$	5.883051268	5.883051268
$\mathcal{T}_2 \cdot \mathcal{T}'_3$	$[(x-z) \cdot \mathcal{V}'_z] [\mathcal{V}_w \cdot (x' - w')]$	-11.795623980	-11.795623980
$\mathcal{T}_3 \cdot \mathcal{T}'_2$	$[\mathcal{V}_z \cdot (x' - z)] [(x-w) \cdot \mathcal{V}_{w'}]$	-18.252744255	-18.252744255
$\mathcal{T}_3 \cdot \mathcal{T}'_3$	$[\mathcal{V}_z \cdot \mathcal{V}'_z] [(x-w) \cdot (x' - w')]$	6.564041350	6.564041350

Here $\mathcal{V}'_z = \frac{(z-w')}{(z-w')^2} + \frac{(x'-z)}{2(x'-z)^2}$ and $\mathcal{V}_{w'} = \frac{(z-w')}{(z-w')^2} - \frac{(x'-w')}{2(x'-w')^2}$. Note in particular that $\mathcal{T}_3 \cdot \mathcal{T}'_3$ **factorises into a pure- w factor times a pure- z factor**; this is the whole story of §6.

3. Transverse UV: there is none

A logarithmic short-distance divergence needs the integrand to go as ρ^{-2} when n points collapse, because n collapsing points supply $\rho^{2(n-1)} \frac{d\rho}{\rho}$. Scanning all $\binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 26$ subsets of $\{x, x', z, w, w'\}$ at $\rho = 10^{-2}, 10^{-4}, 10^{-6}$ (angle-averaged, with the phase in place):

no UV in any of the 26 regions.
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The interesting one is the "loop closing" region $z \rightarrow w$ — the two final-state gluons at the same transverse point. Put $\rho = |z - w|$:

ρ	$\langle F \rangle$	$\langle F \rangle \rho$	$\langle F \rangle \rho^2$
10^{-1}	$3.107773 \cdot 10^0$	0.310777347	$3.107773 \cdot 10^{-2}$
10^{-2}	$3.085291 \cdot 10^1$	0.308529098	$3.085291 \cdot 10^{-3}$
10^{-3}	$3.085068 \cdot 10^2$	0.308506815	$3.085068 \cdot 10^{-4}$
10^{-4}	$3.085066 \cdot 10^3$	0.308506593	$3.085066 \cdot 10^{-5}$
10^{-5}	$3.085066 \cdot 10^4$	0.308506590	$3.085066 \cdot 10^{-6}$

$\langle |F| \rangle \rho$ is flat, so the integrand is ρ^{-1} , not ρ^{-2} : against $d^2\rho \sim \rho d\rho$ this is $\int d\rho$, **linearly convergent, no log**.

Why, structurally. Two things conspire.

- Each $\mathcal{W}^{ji}(x, z, w)$ is only ρ^{-1} singular at $z \rightarrow w$, not ρ^{-2} : the δ^{ji} numerator $A - B = (x - z)^2 - (x - w)^2 = -2(x - w) \cdot (z - w) + (z - w)^2$ vanishes like ρ , cancelling one power of $C = \rho^2$; and $\mathcal{V}_w, \mathcal{V}_z$ are ρ^{-1} .
- Only **one** of the two vertices is singular there. The conjugate vertex is $\mathcal{W}^{ji}(x', z, w')$ — regular at $z = w$ because $w' \neq w$. And $w' \neq w$ is enforced by the measurement itself: the phase $e^{-ik_\perp \cdot (w' - w)}$ is what separates the amplitude and conjugate emission points of the *observed* gluon.

Forcing both to be singular, $z \rightarrow w$ and $w' \rightarrow w$, makes $|F| \sim \rho^{-2}$ (checked: $\langle |F| \rangle \rho^2 = 1.067324 \cdot 10^{-1}$, constant over $\rho = 10^{-2} \dots 10^{-5}$), but the three-point measure is $\rho^3 d\rho$, so it goes as $\int \rho d\rho$: convergent again, and in any case that corner is oscillation-killed once $k_\perp \neq 0$.

Answer to "a loop at the final state must have a UV". It would, if the loop momentum flowed through a single propagator pair. Here the unobserved gluon's transverse position z is *shared* between amplitude and conjugate (its $\mathbf{p}_\perp$ was integrated, setting $z = z'$), and its two kernels attach to *different* points, w on one side and w' on the other. The measurement phase keeps w and w' apart, so the two singular kernels never sit on the same separation. That is the same mechanism that made every other real (cut) row UV-free; the only UV in this calculation lives in the purely virtual row $\mathfrak{A}^{(1)\dagger} \times \mathfrak{A}^{(3)}$, where it comes out as $-2b_0/\epsilon$.

4. Transverse IR: the one genuine divergence, at $|z| \rightarrow \infty$

At large $|z|$ the vertex degenerates. Using $A \rightarrow |z|^2$, $C \rightarrow |z|^2$ in §2,

$$\frac{\mathcal{W}^{ji}(x, z, w)}{S D} \longrightarrow -\frac{1}{2p^+k^+} \frac{(x-z)^j(x-w)^i}{(x-z)^2(x-w)^2}, \quad D \equiv p^+A + k^+B,$$

which is the $\mathcal{T}_2$ (i.e. S/k^+) structure: $\mathcal{V}_w \rightarrow -\frac{(x-w)}{2(x-w)^2}$ while $(x-z)^j$ grows like $|z|$. The $\mathcal{T}_1$ and $\mathcal{T}_3$ structures fall faster. Squaring, $(x-z) \cdot (x'-z)/(AA') \rightarrow 1/|z|^2$ with no angular cancellation, so the integrand has a flat $1/|z|^2$ tail: **a logarithmic IR divergence in z** . Directly, at fixed p^+ ,

$$F \longrightarrow \frac{1}{4p^+k^+} \frac{(x-w) \cdot (x'-w')}{(x-w)^2(x'-w')^2} \frac{1}{|z|^2},$$

confirmed to 9 decimals against the prediction at six values of p^+ . This is a real IR divergence of this row: the colour average does **not** cut it off, because $U(z) \rightarrow 1$ as z leaves the target and $\langle \mathcal{C} \rangle$ tends to a finite non-zero limit.

The other large-distance ends are harmless. $|x|, |x'| \rightarrow \infty$ also look log-divergent in a bare scan, but x, x' are the arguments of ρ : the colour-charge correlator has the transverse support of the target, so those ends are cut at the target radius in the standard way. $|w|, |w'| \rightarrow \infty$ are Fresnel-oscillatory — the scan does not converge there, and it should not: $\int d^2w \frac{(x-w)^i}{(x-w)^2} e^{ik \cdot w}$ is a conditionally convergent Fourier transform, finite distributionally.

5. The p^+ integration, exactly

Only six independent longitudinal weights occur, because the $\mathcal{T}_a$ carry 1, S/k^+ , S/p^+ against the overall p^+k^+/S^2 :

$$h_{11} = \frac{p^+k^+}{S^2}, \quad h_{12} = \frac{p^+}{S}, \quad h_{13} = \frac{k^+}{S}, \quad h_{22} = \frac{p^+}{k^+}, \quad h_{23} = 1, \quad h_{33} = \frac{k^+}{p^+}.$$

The two energy denominators factorise, $D D' = AA' (p^+ + \alpha)(p^+ + \alpha')$ with

$$\alpha \equiv \frac{k^+B}{A} = \frac{k^+(x-w)^2}{(x-z)^2}, \quad \alpha' \equiv \frac{k^+B'}{A'} = \frac{k^+(x'-w')^2}{(x'-z)^2},$$

both strictly positive, so **no pole sits on the contour** — no principal value, no $i\pi$, this row is clean. Define

$$\mathcal{M}_{ab} = \frac{1}{AA'} \int_{\Lambda}^{\vee-k^+} dp^+ \frac{h_{ab}(p^+)}{(p^+ + \alpha)(p^+ + \alpha')} = \frac{1}{AA'} [F_{ab}(\vee - k^+) - F_{ab}(\Lambda)].$$

All six primitives are elementary; each was verified symbolically ($\partial_p F_{ab} - h_{ab}/[(p + \alpha)(p + \alpha')] \equiv 0$) and numerically:

pair	h	quadrature	closed form	diff
$\mathcal{T}_2\mathcal{T}_3'$	1	0.882636709	0.882636709	$4 \cdot 10^{-16}$
$\mathcal{T}_2\mathcal{T}_2'$	p^+/k^+	14.366017631	14.366017631	$2 \cdot 10^{-12}$
$\mathcal{T}_3\mathcal{T}_3'$	k^+/p^+	16.886103148	16.886103148	$1 \cdot 10^{-13}$
$\mathcal{T}_1\mathcal{T}_3'$	k^+/S	0.442719469	0.442719469	$3 \cdot 10^{-14}$
$\mathcal{T}_1\mathcal{T}_2'$	p^+/S	0.439917240	0.439917240	$5 \cdot 10^{-14}$
$\mathcal{T}_1\mathcal{T}_1'$	p^+k^+/S^2	0.144304923	0.144304923	$6 \cdot 10^{-14}$

($k^+ = 1$, $A = 0.8$, $B = 1.7$, $A' = 1.3$, $B' = 0.6$, $\Lambda = 10^{-8}$, $\vee = 10^7$.)

5.1 Where the endpoint logs live

Expanding the primitives at both ends:

- $\log \frac{1}{\Lambda}$ occurs **only** in $\mathcal{M}_{33}$, with residue $\frac{1}{k^+ BB'}$. This is the rapidity divergence.
- $\log \vee$ occurs **only** in $\mathcal{M}_{22}$, with residue $\frac{1}{k^+ AA'}$. It is not a divergence: $\vee = q^+$ is fixed by kinematics.
- $\mathcal{M}_{11}, \mathcal{M}_{12}, \mathcal{M}_{13}, \mathcal{M}_{23}$ are finite at both ends.

Taking $\Lambda \rightarrow 0$ and $\vee \rightarrow \infty$ in the finite parts, and writing everything in A, B, A', B' (all four verified against the exact integrals to 10^{-9}):

$$\begin{aligned}\mathcal{M}_{23} &= \frac{1}{k^+ \Delta} \log \frac{AB'}{A'B}, \quad \Delta \equiv AB' - A'B, \\ \mathcal{M}_{22} &= \frac{1}{k^+ AA'} \left[\log(\vee - k^+) - \frac{AB' \log \frac{k^+ B'}{A'} - A'B \log \frac{k^+ B}{A}}{\Delta} \right], \\ \mathcal{M}_{33} &= \frac{1}{k^+} \left[\frac{1}{BB'} \log \frac{1}{\Lambda} + \frac{1}{\Delta} \left(\frac{A}{B} \log \frac{k^+ B}{A} - \frac{A'}{B'} \log \frac{k^+ B'}{A'} \right) \right], \\ \mathcal{M}_{13} &= -\frac{k^+}{AA'} \left[\frac{\log k^+}{(\alpha - k^+)(\alpha' - k^+)} + \frac{\log \alpha}{(k^+ - \alpha)(\alpha' - \alpha)} + \frac{\log \alpha'}{(k^+ - \alpha')(\alpha - \alpha')} \right], \\ \mathcal{M}_{12} &= \frac{1}{AA'} \left[\frac{k^+ \log k^+}{(\alpha - k^+)(\alpha' - k^+)} + \frac{\alpha \log \alpha}{(k^+ - \alpha)(\alpha' - \alpha)} + \frac{\alpha' \log \alpha'}{(k^+ - \alpha')(\alpha - \alpha')} \right], \\ \mathcal{M}_{11} &= -\frac{k^+}{AA'} \left[\frac{1}{(\alpha - k^+)(\alpha' - k^+)} + \frac{(\alpha \alpha' - k^{+2}) \log k^+}{(\alpha - k^+)^2 (\alpha' - k^+)^2} - \frac{\alpha \log \alpha}{(k^+ - \alpha)^2 (\alpha' - \alpha)} - \frac{\alpha' \log \alpha'}{(k^+ - \alpha')^2 (\alpha - \alpha')} \right].\end{aligned}$$

The apparent poles at $\alpha = k^+, \alpha' = k^+, \alpha = \alpha'$ (and $\Delta = 0$) are all **removable** — each bracket is regular there, as the partial-fraction construction guarantees.

5.2 The assembled result

$$\begin{aligned}N_2 &= \frac{g^4}{32\pi^7} \int d^2x d^2x' d^2z d^2w d^2w' e^{-ik_\perp \cdot (w' - w)} \langle \mathcal{C} \rangle \mathcal{I}(x, x', z, w, w') \\ \mathcal{I} &= \frac{(A - B)(A' - B')}{2CC'} \mathcal{M}_{11} + \left[\frac{A - B}{2C} (x' - z) \cdot \mathcal{V}_{w'} + \frac{A' - B'}{2C'} (x - z) \cdot \mathcal{V}_w \right] \mathcal{M}_{12} \\ &+ \left[\frac{A - B}{2C} \mathcal{V}'_z \cdot (x' - w') + \frac{A' - B'}{2C'} \mathcal{V}_z \cdot (x - w) \right] \mathcal{M}_{13} + [(x - z) \cdot (x' - z)] [\mathcal{V}_w \cdot \mathcal{V}_{w'}] \mathcal{M}_{22} \\ &+ \left[[(x - z) \cdot \mathcal{V}'_z] [\mathcal{V}_w \cdot (x' - w')] + [(x' - z) \cdot \mathcal{V}_z] [\mathcal{V}_{w'} \cdot (x - w)] \right] \mathcal{M}_{23} + [\mathcal{V}_z \cdot \mathcal{V}'_z] [(x - w) \cdot (x' - w')] \mathcal{M}_{33}\end{aligned}$$

with $\langle \mathcal{C} \rangle = \langle [U^{ac}(w) f^{cbd} \rho^d(x) - U^{bc}(z) f^{acd} U^{de}(x) \rho^e(x)] [\cdot]^\dagger_{x', w'} \rangle$ and the $\mathcal{M}_{ab}$ above.

6. The transverse log after the p^+ integration

This is the prettiest part. Take $|z| \rightarrow \infty$ in $\mathcal{I}$ *after* integrating p^+ . Four pieces survive at order $|z|^{-2}$ — $\mathcal{M}_{22}, \mathcal{M}_{33}, \mathcal{M}_{23}$ (twice) — and they conspire:

- $\mathcal{M}_{22}$ carries $+\log |z|^2$ from its Λ -end, $\mathcal{M}_{33}$ carries $-\log |z|^2$: **the $\log^2$ cancels**, leaving a single log.
- The leftovers $\mp \frac{B+B'}{B'-B} \log \frac{B'}{B}$ of $\mathcal{M}_{22} + \mathcal{M}_{33}$ cancel exactly against $\mathcal{M}_{23} + \mathcal{M}_{32}$.

What survives is

$$\mathcal{I} \xrightarrow{|z| \rightarrow \infty} \frac{1}{4k^+} \log \frac{\vee - k^+}{\Lambda} \frac{(x - w) \cdot (x' - w')}{(x - w)^2 (x' - w')^2} \frac{1}{|z|^2} + O(|z|^{-3}).$$

Checked against the fully assembled $\mathcal{I}$ at four choices of $(\Lambda, \vee)$ and four radii:

Λ	$\vee$	$ z = 10^3$	10^4	10^5	10^6	predicted
10^{-6}	10^5	0.6137632	0.6137598	0.6137598	0.6137598	0.6137598
10^{-9}	10^5	0.7811545	0.7811489	0.7811489	0.7811489	0.7811489
10^{-6}	10^8	0.7811559	0.7811492	0.7811491	0.7811491	0.7811491
10^{-3}	10^3	0.3347556	0.3347540	0.3347539	0.3347539	0.3347539

(entries are $\langle \mathcal{I} \rangle_{\hat{z}} |z|^2$.)

Read it off. The transverse factor $\frac{(x-w) \cdot (x'-w')}{(x-w)^2 (x'-w')^2}$ is precisely the **LO** single-gluon-production structure: the Weizsäcker–Williams kernel for the observed gluon, emitted from x at w in the amplitude and from x' at w' in the conjugate. The unobserved gluon has **completely decoupled** — all that is left of it is $\int d^2 z / |z|^2$ and the rapidity span $\log \frac{\vee - k^+}{\Lambda}$. That is the definition of the eikonal/soft region. The $\log \frac{1}{\Lambda}$ is the JIMWLK rapidity divergence, to be absorbed into the small- x evolution of the target; the accompanying $\int d^2 z / |z|^2$ is cut off only once the partner terms are included — the missing $(2\rho) \times (2\rho)$ piece, the N_1 cross terms (whose $|z| \rightarrow \infty$ plateau I checked is exactly $-2 \times$ this one), and the virtual rows, where the same eikonal kernel appears with the opposite sign. Taken by itself, N_2 is log–IR divergent in $|z|$, and that is expected, not an error.

7. DGLAP: no — and here is why, three independent ways

7.1 There is no collinear region

A final-state (fragmentation) collinear log has to come from the two produced gluons approaching each other in transverse space, i.e. $|z - w| \rightarrow 0$. §3 shows that region is ρ^{-1} , giving $\int d\rho$: **convergent, no log**. The reason is the measurement itself: the conjugate vertex sits at $w' \neq w$ and is regular there, and $e^{-ik_\perp \cdot (w' - w)}$ is exactly what prevents w' from following w . No collinear region, no collinear log, no fragmentation function.

7.2 The splitting function is not there

Test the ζ -dependence of the one log that does exist. Hold $S = p^+ + k^+$ fixed and vary $\zeta = k^+ / S$; strip the transverse factor:

ζ	plateau $\times 4S^2$	$1/(\zeta\bar{\zeta})$	$C_{UV}(\zeta) = \frac{P_{gg}}{2N_c}$	ratio
0.05	21.0526313	21.0526316	19.100132	1.102224
0.15	7.8431372	7.8431373	5.970637	1.313618
0.30	4.7619047	4.7619048	2.971905	1.602307
0.50	4.0000000	4.0000000	2.250000	1.777778
0.70	4.7619047	4.7619048	2.971905	1.602307
0.90	11.1111111	11.1111111	9.201111	1.207584
0.97	34.3642610	34.3642612	32.393361	1.060843

The weight is **exactly** $1/(\zeta\bar{\zeta})$ — to nine digits, at every ζ . Now

$$\frac{P_{gg}(\zeta)}{2N_c} = C_{UV}(\zeta) = \frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + \zeta\bar{\zeta} = \frac{1}{\zeta\bar{\zeta}} + \zeta\bar{\zeta} - 2.$$

So this row supplies the $\frac{1}{\zeta\bar{\zeta}} = \frac{\zeta}{\bar{\zeta}} + \frac{\bar{\zeta}}{\zeta} + 2$ part — **both soft ends** — and none of the regular $\zeta\bar{\zeta} - 2$ part. The ratio to C_{UV} runs from 1.06 to 1.78: it is not P_{gg} . A DGLAP log must carry the complete splitting function; a purely soft $1/(\zeta\bar{\zeta})$ weight is the signature of eikonal emission.

7.3 The kinematics are wrong for fragmentation

In the row with two $\mathfrak{C}$ vertices — the one that *does* fragment — the $\mathfrak{C}$ delta function ties the emission points together as $w' - w = \frac{S}{k^+}(y' - y) = \frac{1}{\zeta}(y' - y)$, so the phase becomes $e^{-i(k_\perp/\zeta) \cdot (y' - y)}$: the hard factor is evaluated at the **parent's** momentum $k_\perp/\zeta$, and the longitudinal measure comes out as $\frac{d\zeta}{\zeta^2}$. Those are the two hallmarks of a fragmentation function.

Here nothing of the sort happens: w and w' are free integration variables, and the phase is $e^{-ik_{\perp} \cdot (w' - w)}$ with the **observed** $k_{\perp}$, unrescaled. There is no ξ -integral left against the log either — after §5 the log's coefficient is the pure number $\log \frac{\sqrt{-k^+}}{\Lambda}$, i.e. $\log \frac{1}{x_g}$, not $\int \frac{d\xi}{\xi} P_{gg}(\xi) \log \mu^2$.

Verdict on DGLAP. N_2 produces a small- x (JIMWLK/BK) rapidity log multiplying the LO structure, not a collinear DGLAP log. Two final-state gluons with one integrated out is a necessary condition for fragmentation, not a sufficient one — you also need the two kernels to become singular on the *same* separation, and the measurement phase forbids that here. The DGLAP/fragmentation logs in this calculation live in the rows where a $\mathfrak{C}$ vertex ties the observed gluon's amplitude and conjugate positions together.

8. What to do next

1. Fix the three bookkeeping items of §1 (Wilson-line transpose, the $\int_{x,x'}$, the missing $(2\rho) \times (2\rho)$ piece).
2. Use §5.2 as the p^+ -integrated answer. No $+$ prescription is needed or appropriate: there is no pole on the p^+ contour ($\alpha, \alpha' > 0$), and the only endpoint singularity is the $\log \frac{1}{\Lambda}$ of $\mathcal{M}_{33}$, which is the rapidity divergence and belongs in the evolution kernel, not in a plus-distribution.
3. Combine the $|z| \rightarrow \infty$ log of §6 with the same tail in N_1 , in the missing 4ρ piece, and in the virtual rows before quoting anything finite.

All statements above are backed by `src/row17_check.py`, `src/row17_div.py`, `src/row17_final.py`, `src/row17_zlog.py`, `src/row17_dglap.py`, with raw output in `RESULTS_row17.txt`.